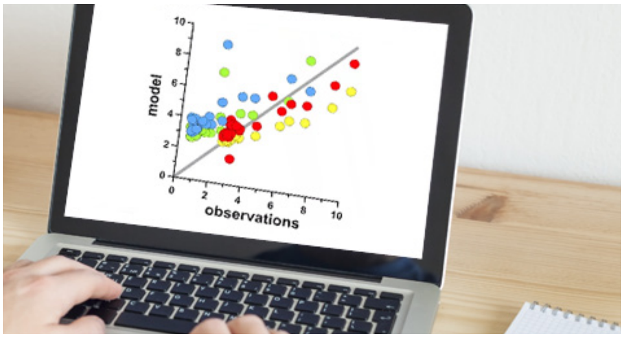
R Series: Correlation

This sixteenth article in the R series will introduce you to correlation.



In this article, we shall explore correlation. We will use R version 4.2.1 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version

R version 4.2.1 (2022-06-23) -- "Funny-Looking Kid"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

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```

The *cor()* function in R can compute the correlation between two sets of vectors. Its usage is as follows:

```
cor(x, y, na.rm, use, method)
```

The correlation function accepts the following arguments:

Argument	Description
x	numeric vector, matrix or data frame
y	vector or NULL
na.rm	logical value to remove missing values
use	string to specify the computing method
method	'pearson', 'kendall', or 'spearman' coefficient

Let us create three vectors 'x', 'y' and 'z' for comparison using the *sin()* and *cos()* functions as follows:

```
> t = seq(0, 10, 0.1)
> x = sin(t)
> y = sin(t + 0.05)
> z = cos(t)
```

The first few values of the vectors are listed below:

```
> head(x)
[1] 0.00000000 0.09983342 0.19866933 0.29552021 0.38941834
0.47942554

> head(y)
[1] 0.04997917 0.14943813 0.24740396 0.34289781 0.43496553
0.52268723

> head(z)
[1] 1.0000000 0.9950042 0.9800666 0.9553365 0.9210610
0.8775826
```

The correlation between the 'x' and 'y' vectors as well as the 'x' and 'z' vectors is shown below:

```
> cor(x, y)
[1] 0.9985339

> cor (x, z)
[1] 0.05483627
```

We observe that there is a high correlation of 0.99 between the 'x' and 'y' vectors as they come from the same sine function. The 'x' and 'z' vectors have a low correlation of 0.05 as they are defined using sine and cosine functions respectively.

Consider the *mtcars* data set available in the *lattice* library:

```
> head(mtcars)
             mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4   21.0    6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag 21.0  6 160 110 3.90 2.875 17.02 0  1    4    4
Datsun 710  22.8  4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive  21.4  6  258 110 3.08 3.215 19.44 1 0  3    1
Hornet Sportabout 18.7 8  360 175 3.15 3.440 17.02 0 0  3   2
Valiant     18.1   6  225 105 2.76 3.460 20.22  1  0  3    1
```